Learning Objective 1.1: I can define noise figure and cascade it through a receiver chain to predict system noise performance.
Learning Objective 1.2: I can compute a link budget and predict received SNR for a specified geometry, transmitter, and antenna configuration.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Decibels and the ISM bands.** A small 915 MHz telemetry radio puts out 100 mW. It connects to a 9 dBi antenna through a cable and connectors with 2 dB of loss.

   (a) Express 5 W and 0.1 μW in dBm.

   $$10\log_{10}\left(\frac{5000\text{ mW}}{1\text{ mW}}\right) = 37.0\text{ dBm}$$
   $$10\log_{10}\left(\frac{10^{-4}\text{ mW}}{1\text{ mW}}\right) = -40.0\text{ dBm}$$

   5 W = 37.0 dBm

   0.1 μW = -40.0 dBm

   (b) Find the radio's EIRP in dBm and in milliwatts.

   $$\text{EIRP} = 10\log_{10}(100) - 2 + 9 = 20 - 2 + 9 = 27\text{ dBm}$$
   $$= 10^{27/10} = 501\text{ mW}$$

   EIRP = 27 dBm

   EIRP = 501 mW

   (c) Which of the course SDRs (RTL-SDR, PlutoSDR, HackRF One) can receive the 2.4 GHz ISM band, and which can transmit in it?

   The PlutoSDR (325 MHz to 3.8 GHz) and the HackRF One (1 MHz to 6 GHz) both cover 2.4 GHz and can both transmit. The RTL-SDR tops out near 1.77 GHz and is receive-only, so it can reach the 915 MHz band but not 2.4 GHz.

   (d) Explain in one or two sentences why Wi-Fi, Bluetooth, and most hobby drones all operate in the same few bands.

   The ISM bands are the bands where FCC Part 15 allows low-power transmitters to operate without a license, as long as they accept interference from other users. Every consumer device that cannot require its user to hold a license ends up there.

2. **Free-space path loss.** A 915 MHz radio talks to a ground station 10 km away. Both antennas have 6 dBi of gain, and the transmitter delivers 1 W to its antenna.

   (a) Find the free-space path loss at 915 MHz and 10 km.

   $$L_p = 20\log_{10}\left(915 \times 10^6\right) + 20\log_{10}\left(10^4\right) - 147.55$$
   $$= 179.2 + 80.0 - 147.55 = 111.7 \text{ dB}$$

   L_p = 111.7 dB

   (b) Find the received power in dBm.

   $$P_r = P_t + G_t + G_r - L_p$$
   $$= 30 + 6 + 6 - 111.7 = -69.7 \text{ dBm}$$

   P_r = −69.7 dBm

   (c) The same system moves to 2.45 GHz with the same antenna gains and twice the distance. Find the new path loss using the rules of thumb, then check it with the full expression.

   Doubling the distance adds 6.0 dB. The frequency ratio adds $20\log_{10}(2450/915) = 8.6$ dB. So

   $$L_p \approx 111.7 + 6.0 + 8.6 = 126.3 \text{ dB}$$
   $$L_p = 20\log_{10}\left(2.45 \times 10^9\right) + 20\log_{10}\left(2 \times 10^4\right)$$
   $$- 147.55 = 126.3 \text{ dB}$$

   L_p = 126.3 dB

   (d) Path loss is not absorption, yet it grows with frequency. Explain why in one or two sentences.

   The power spreads over a sphere regardless of frequency, but an antenna of fixed gain has an effective area proportional to λ^2. At a higher frequency the same gain collects power over a smaller area, so less of the spreading wave reaches the receiver.

3. **The noise floor.** An RTL-SDR with a 6 dB noise figure receives a broadcast FM station. One FM channel occupies about 200 kHz.

   (a) Find the receiver's noise floor in one 200 kHz channel.

   $$N = -174 + 10\log_{10}\left(200 \times 10^3\right) + 6$$
   $$= -174 + 53.0 + 6 = -115.0 \text{ dBm}$$

   N = -115.0 dBm

   (b) The station arrives at -80 dBm. Find the SNR in the channel.

   $$\text{SNR} = -80 - (-115.0) = 35.0 \text{ dB}$$

   SNR = 35.0 dB

   (c) A student leaves the receiver filter open to the full 2.4 MHz sample rate and measures the station's power against all of that noise. Find the noise floor and the SNR the student would compute.

   $$N = -174 + 10\log_{10}\left(2.4 \times 10^6\right) + 6 = -104.2 \text{ dBm}$$
   $$\text{SNR} = -80 - (-104.2) = 24.2 \text{ dB}$$

   N = -104.2 dBm

   SNR = 24.2 dB

   (d) Explain why the answers to (b) and (c) differ, and which one the FM demodulator experiences.

   Noise power grows with bandwidth, but the station's power does not: it all lies inside its 200 kHz channel. Letting in 12 times the bandwidth adds 10.8 dB of noise and no signal. A demodulator that filters to the channel first sees the 35.0 dB of part (b), which is why every receiver filters before it demodulates.

4. **Cascaded noise figure.** A ground station has a 2 dB cable run, an LNA with NF 0.8 dB and 18 dB of gain, and an SDR with NF 7 dB.

   (a) Convert each stage to a linear noise factor and gain.

   Cable: $F = 10^{0.2} = 1.585$, $G = 10^{-0.2} = 0.631$. LNA: $F = 10^{0.08} = 1.202$, $G = 10^{1.8} = 63.1$. SDR: $F = 10^{0.7} = 5.01$.

   F_{cable} = 1.585

   F_{LNA} = 1.202

   (b) The cable runs from the antenna to the LNA, which sits next to the SDR. Find the system noise figure.

   $$F = 1.585 + \frac{1.202 - 1}{0.631} + \frac{5.01 - 1}{0.631 \times 63.1}$$
   $$= 1.585 + 0.320 + 0.101 = 2.006$$
   $$\text{NF} = 10\log_{10}(2.006) = 3.0 \text{ dB}$$

   NF = 3.0 dB

   (c) The LNA moves to the antenna, ahead of the cable. Find the new system noise figure.

   $$F = 1.202 + \frac{1.585 - 1}{63.1} + \frac{5.01 - 1}{63.1 \times 0.631}$$
   $$= 1.202 + 0.009 + 0.101 = 1.312$$
   $$\text{NF} = 10\log_{10}(1.312) = 1.2 \text{ dB}$$

   NF = 1.2 dB

   (d) Moving the LNA lowered the noise figure by 1.8 dB. Explain why the cable's loss matters so much more in front of the LNA than behind it.

   In front of the LNA the cable is the first stage, so its full noise factor counts and nothing divides it down: its 2 dB adds almost directly to the system noise figure. Behind the LNA, the LNA's gain of 63 divides down the cable's excess noise, and the cable adds only about 0.2 dB.

5. **A GPS link budget.** A GPS satellite transmits the L1 signal at 1575.42 MHz with an EIRP of 27 dBW. A receiver on the ground sees the satellite directly overhead at 20,200 km through a 0 dBi antenna. Polarization and atmospheric losses total 2 dB. The receiver has a 2 dB noise figure and a 2 MHz bandwidth.

   (a) Find the free-space path loss and the received power in dBm.

$$\begin{aligned} L_p &= 20\log_{10}\left(1.57542\times10^{9}\right) \\ &\quad + 20\log_{10}\left(2.02\times10^{7}\right) - 147.55 \\ &= 183.9 + 146.1 - 147.55 = 182.5 \text{ dB} \\ \text{EIRP} &= 27 \text{ dBW} = 57 \text{ dBm} \\ P_r &= 57 - 182.5 + 0 - 2 = -127.5 \text{ dBm} \end{aligned}$$

L_p = 182.5 dB

P_r = −127.5 dBm

   (b) Find the noise floor and the SNR at the receiver.

$$\begin{aligned} N &= -174 + 10\log_{10}\left(2\times10^{6}\right) + 2 = -109.0 \text{ dBm} \\ \text{SNR} &= -127.5 - (-109.0) = -18.5 \text{ dB} \end{aligned}$$

N = −109.0 dBm

SNR = −18.5 dB

   (c) The receiver correlates the signal against the satellite's known 1023-chip code, which raises the SNR by $10\log_{10}(1023)$. Find the SNR after correlation.

$$\begin{aligned} 10\log_{10}(1023) &= 30.1 \text{ dB} \\ \text{SNR}_{\text{corr}} &= -18.5 + 30.1 = 11.6 \text{ dB} \end{aligned}$$

SNR = 11.6 dB

   (d) A car's GPS antenna sits on the roof with 5 m of cable to the dashboard receiver. Where should the LNA go, and why? Use your answer to (b) to explain what is at stake.

   The LNA belongs at the antenna, ahead of the cable; that is why GPS antennas with a built-in LNA, active antennas, are common. The signal already arrives 18.5 dB below the noise floor, and the correlation leaves only about 12 dB of SNR, so every dB of cable loss in front of the first amplifier would come straight out of that.

6. **Reading SNR on an SDR.** A GNU Radio frequency sink shows an unmodulated carrier from a signal generator, received by an RTL-SDR running at $f_s = 2.4$ MHz with an FFT size of 2048. The carrier's peak reads -32 dBFS and the noise floor beside it reads -78 dBFS.

   (a) Find the SNR the display shows, and the width of one FFT bin.

   $$\text{SNR} = -32 - (-78) = 46 \text{ dB}$$
   $$\frac{f_s}{N} = \frac{2.4 \text{ MHz}}{2048} = 1.17 \text{ kHz}$$

   SNR = 46 dB

   bin = 1.17 kHz

   (b) The display reads dBFS, not dBm. Why can you still read an SNR off it?

   SNR is a difference between two levels on the same axis, so the unknown offset between dBFS and dBm cancels.

   (c) You change the FFT size from 2048 to 8192. What happens to the noise floor, to the carrier's peak, and to the SNR the display shows?

   Each bin is four times narrower and holds a quarter of the noise, so the floor drops by about 6 dB, 3 dB for each doubling. The carrier's power is all in one bin, so its peak stays at -32 dBFS, and the displayed SNR rises to about 52 dB. Nothing about the signal changed; only the display did.

   (d) Back at 2048 points, the receiver filters to 2 kHz before it demodulates. Find the SNR in that bandwidth.

   Scale the floor from one bin to 2 kHz:

   $$10\log_{10}\left(\frac{2000}{1172}\right) = 2.3 \text{ dB}$$
   $$\text{SNR} = 46 - 2.3 = 43.7 \text{ dB}$$

   SNR = 43.7 dB

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